Sri Lanka Institute of Information Technology

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SE4031 - Games Development

Assignment 02

Elephant Jungle Survival

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1. Summary of the Game Concept

Elephant Jungle Survival is a 2D endless survival game set in a jungle environment. In this game, the player controls a young boy who is trying to escape from an angry elephant chasing him. The main goal of the game is to survive as long as possible while avoiding obstacles, collecting items, and dealing with enemies.

The game is played on a continuously moving path where different objects appear, such as obstacles, enemies, and health items. The player must move carefully to avoid hitting obstacles, as collisions will reduce the player's health. If the health reaches zero, the game ends. To help the player survive longer, health packs (bananas) are placed along the path. When collected, these increase the player's health. The player can also shoot projectiles using the mouse to destroy enemies. This adds an action element to the game and allows the player to gain more score. The score increases based on how long the player survives and how many enemies are defeated. As the game continues, the player's speed gradually increases, making the game more challenging over time. This requires the player to react faster and improves the overall difficulty of the game.

The game uses a cartoon-style jungle theme with simple and colourful visuals. Sound effects and background music are included to make the gameplay more engaging. Overall, **Elephant Jungle Survival** is a fast-paced survival game that combines movement, shooting, and health management, creating an enjoyable and challenging experience for the player.

2. Core Features Implementation

1. Health System

The player has a health system that is displayed as a numeric value on the top-left corner of the screen.

- The player starts with a maximum health value of 100.
- Health decreases when:
  - The player collides with obstacles (such as wooden logs)
  - The player gets hit by enemies (elephant)
- Even after collision, the player continues moving, but the health value is reduced.
- When the health reaches 0, the game stops and a Game Over screen is displayed.
- When the player restarts the game, the health is reset back to the maximum value.

This system ensures that the player can make small mistakes but still continue playing until health is fully depleted.

2. Score System

The game includes a score system that tracks the player's performance.

- The score is displayed on the top-right corner of the screen.

- The score increases when:
    - The player survives for a longer time
    - The player destroys enemies using projectiles
- The score system encourages the player to survive longer and actively engage with enemies.
- When the game is restarted, the score resets to **0**.

3. Health Packs

Health packs are implemented as collectible banana objects.

- Bananas appear randomly along the path.
- When the player collides with a banana:
    - The banana is immediately destroyed
    - The player's health increases
- Health cannot exceed the maximum limit (100).
- If the player does not collect the banana and moves past it, the banana is automatically removed after a short time.

This feature helps balance the difficulty by giving players a chance to recover health.

4. Projectile Attack System

The player has the ability to shoot projectiles to interact with enemies.

- When the player clicks the left mouse button, a projectile is created.
- The projectile:
    - Spawns at the player's position
    - Moves forward in the game direction
    - Is destroyed when it hits an enemy or obstacle
    - Is also destroyed automatically after a short time if no collision occurs
- When a projectile hits an enemy:
    - The enemy is destroyed
    - The player gains score

This feature adds an action element to the game and allows players to actively defend themselves.

5. Enemy System

Enemies are implemented to create challenge and interaction in the game.

- The main enemy is the elephant, which is clearly different from health packs.
- Enemy behavior:
    - Moves towards or along the player's path
    - Causes damage to the player on collision
- When the player collides with an enemy:
    - Player health decreases
- When the enemy is hit by a projectile:

    - The enemy is destroyed
    - The player receives score
- Enemies are also removed if they move past the player after some time.

This system increases difficulty and forces the player to react quickly.

6. Speed Increase Mechanic

To make the game progressively challenging, the player speed increases over time.

- The speed gradually increases based on gameplay duration.
- This makes:
    - Obstacles harder to avoid
    - Enemy interactions more difficult
- The speed increase is controlled to ensure the game remains playable and not too difficult too quickly.

This feature keeps the game engaging and prevents it from becoming repetitive.

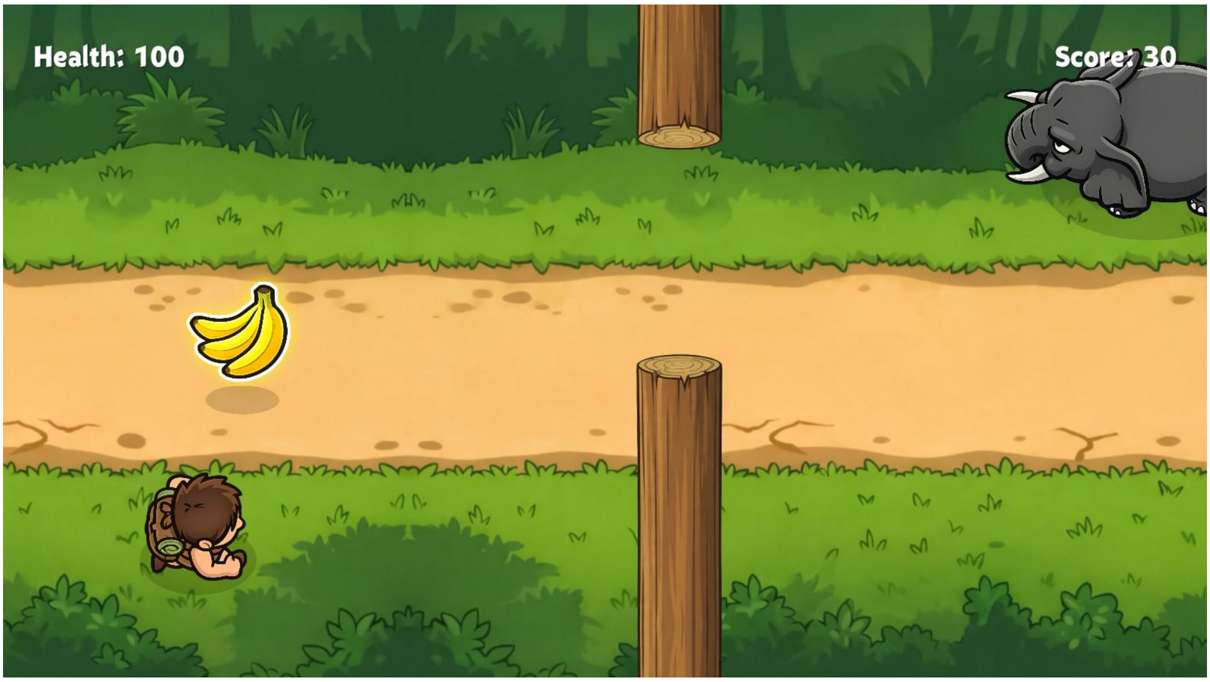
7. Game Over & Restart System

- When the player's health reaches zero:
    - The game stops
    - A Game Over screen appears
- The player can:
    - Restart the game using the Restart button
    - Return to the main menu
- Restarting the game resets:
    - Health
    - Score
    - Game speed

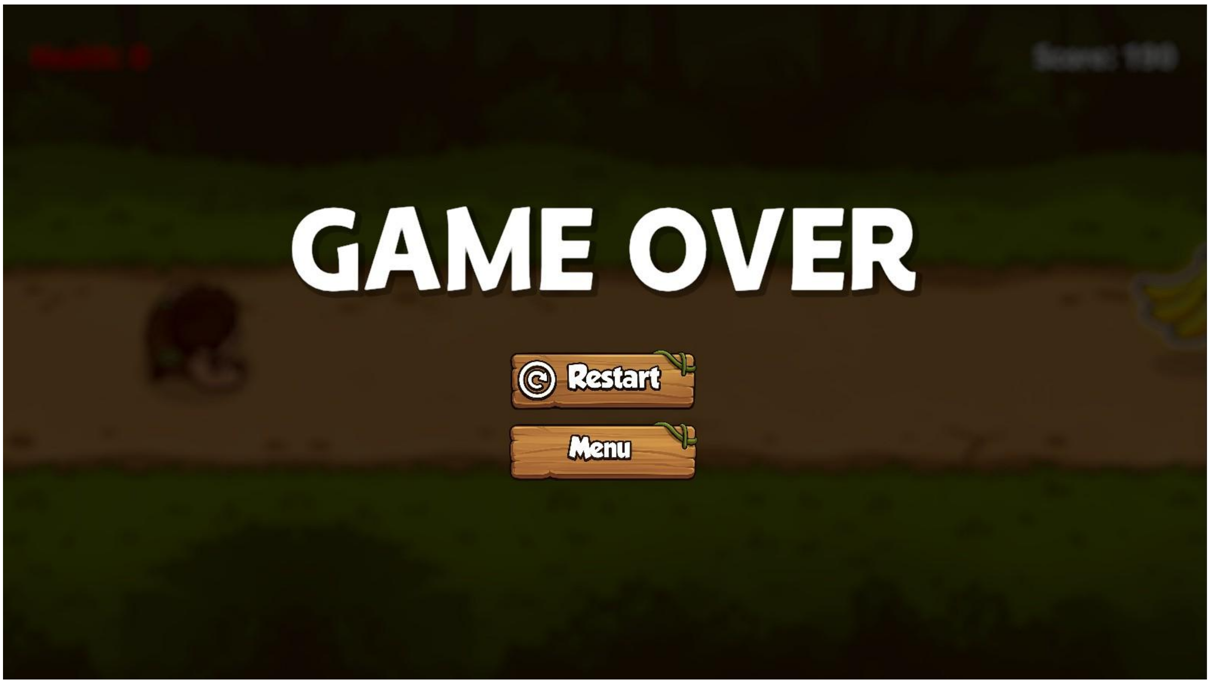
3. Gameplay Screenshots

1. Main Menu

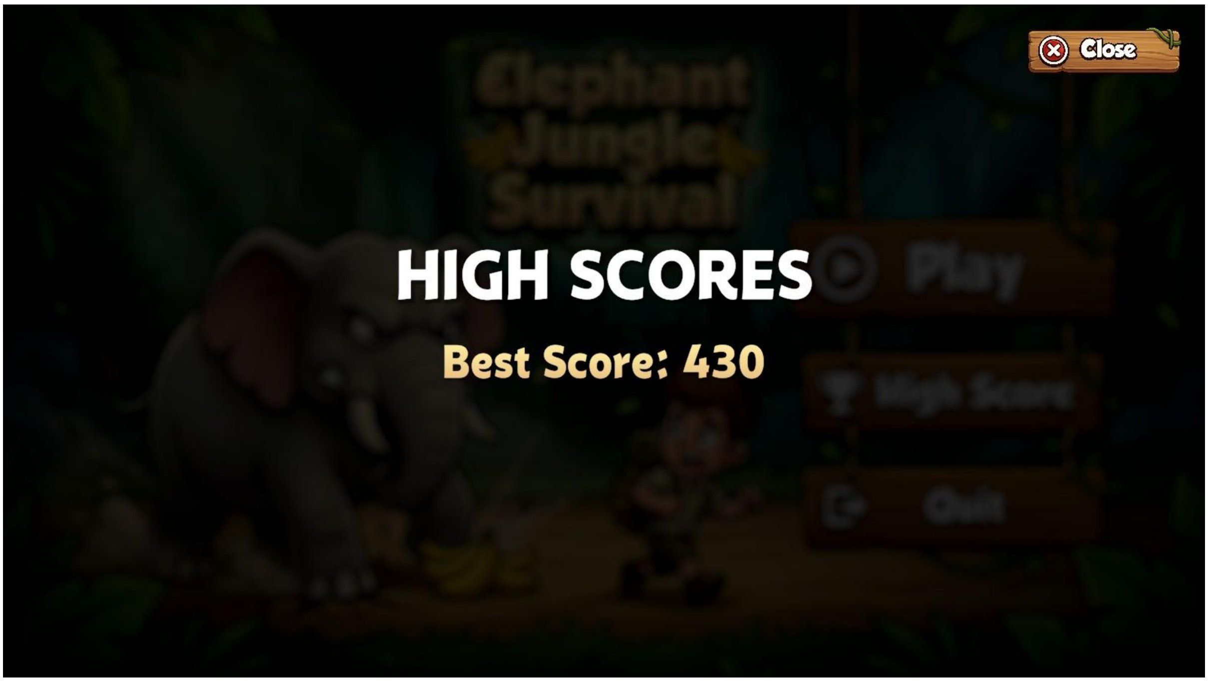
2. Gameplay



3. **Game Over Screen**



4. **High Score Screen**



4. Control Guide

Action	Input
Move Player	Arrow Keys / WASD
Shoot	Left Mouse Click
Start Game	Mouse Click (UI Button)
Restart Game	Mouse Click
Open Menu	Mouse Click

5. Description of the Creative Feature

One of the main creative features implemented in Elephant Jungle Survival is the combination of a story-based chase mechanic with a polished user interface and immersive game design.

1. Designed Game UI and Menu System

A custom jungle-themed user interface was created to improve the visual quality of the game.

- The main menu includes:
    - Play button
    - High Score button
    - Quit button
- The buttons are designed using a wooden jungle style to match the environment.
- A Game Over screen with Restart and Menu options is implemented.
- A High Score panel shows the best score achieved by the player.

This makes the game feel complete and professional rather than a basic prototype.

2. Sound Effects and Background Music

Audio was added to improve player experience and feedback.

- Shooting sound when the player fires a projectile
- Hit sound when interacting with enemies
- Button click sounds for UI interaction
- Game over sound when the player loses
- Background jungle music that plays during gameplay

These sounds make the game more immersive and responsive.

3. Consistent Jungle Theme Design

The game uses a consistent visual style across all elements.

- Jungle background with layered depth
- Cartoon-style characters (player and elephant)
- Matching UI elements (buttons, panels)
- Banana collectibles that fit the theme

This consistency improves the overall look and feel of the game.

4. Smooth Game Flow Experience

The game includes a complete flow from start to finish:

- Main Menu → Gameplay → Game Over → Restart/Menu
- Score tracking and display
- High score system

This ensures the game is fully playable and user-friendly.

6. Credits for External Assets / Tools Used

1. Image and Asset Generation

- **ChatGPT (AI Image Generation)** - Used to generate custom game assets such as UI elements, backgrounds, and design ideas.
- **Nano Banana AI** - Used to generate additional visual assets and improve the visual style of the game.

2. Audio Resources

- **Pixabay** - Used to download free sound effects and background music. All audio assets were used under Pixabay's free license.

3. Development Tools

- **Unity Editor** - Used for scene creation, UI design, and testing the game.
- **Visual Studio** - Used for writing and editing C# scripts.

7. Video Demonstration

Video Link: Gameplay Demo Video.mp4